



Performance and Symptom Validity Testing: Terminology, Conceptualization, Practice Standards, and Future Directions

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Received: 26 May 2026 / Accepted: 14 September 2026
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Abstract

Purpose of Review This commentary summarizes terminology and practice standards for assessing the validity of test scores obtained during neuropsychological evaluations. It also highlights conceptual gaps and emerging directions for improving validity assessment.

Recent Findings The process of assessing the accuracy and interpretability of psychometric test scores is foundational to neuropsychology. However, terminology used to communicate these findings requires conceptual refinement to align with evolving practice standards and advances in psychometric theory. Such terminology must also reflect that validity problems are not reduced to poor effort or volitional exaggeration. Validity is a broader interpretive concern that applies to all practicing neuropsychologists, who must distinguish between presumed causes and patterns of invalidity, while also considering the intended use of scores relative to the patient and context. Symptom- and performance-based scores serve different interpretive purposes, and their validity should be evaluated separately. The strongest validity determinations are based on converging evidence across multiple complementary psychometric- and clinically-based indicators, with attention to base rates of invalid responding and genuinely low scores.

Summary Digital technologies may help the field keep pace with evolving standards and conceptual developments. Technologies can improve validity assessment by capturing more data, analyzing it with advanced algorithms, and doing so in ways that are more accessible for patients and efficient for clinicians. However, neuropsychologists must remain stewards of these digital methods, ensuring their development and use preserve clinical nuance, empirical rigor, and ethical standards. Used appropriately, digital tools may complement rather than replace the uniquely human aspects of validity assessment.

Opinion Statement

Performance and symptom validity assessment is an evidence-based process used to determine whether neuropsychological test findings accurately reflect a person's cognitive abilities or symptomatic experiences. However, terminology in this area has often been used inconsistently, blurring important distinctions among what validity measures assess, what invalid responding implies, and the factors that contribute to invalid responding. In this article, we clarify these distinctions by reviewing terminology used to describe validity assessment methods, patterns of invalid responding, and factors that may threaten validity. We then summarize best-practice standards for performance and symptom validity assessment, emphasizing the use of multiple psychometric and clinically based indicators to estimate the extent to which test scores reflect the constructs they are intended to represent. We explain why such validity findings must be interpreted in relation to the individual, the referral question, and the evaluation context, rather than being treated as sole or direct indicators of non-credible responding or its presumed causes. Finally, we explain how emerging technologies may improve this assessment

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process by capturing and analyzing additional information underlying a person's invalid responses. We further describe how such technologies may help distinguish between the presence and appearance of invalid responding and its presumed causes. Ultimately, we contend that, as long as neuropsychologists remain stewards of these innovations, digital technologies can improve validity assessment while preserving the central role of clinical judgment.

Keywords Performance validity tests · Symptom validity tests · Malingering · Neuropsychological testing · Effort testing · Feigning

Introduction

Defining Validity

Research on performance and symptom validity has evolved from early medicolegal concerns about malingering into a broader, evidence-based approach for evaluating the accuracy and interpretability of neuropsychological test scores. This shift is reflected in the terminology itself. What was once referred to as “effort testing,” and later as “symptom validity testing,” is now called “performance and symptom validity assessment.” These terms are now ubiquitous in neuropsychology, but they still require conceptual development.

As described by Larrabee [1] and other pioneers within this field [2], performance and symptom validity indicate whether scores obtained during psychometric testing provide an accurate and interpretable representation of one's neuropsychological abilities or experiences. In this sense, validity assessment may pertain to the broader issue of construct validity, as neuropsychological test scores can only provide meaningful inferences when they represent the constructs they are designed to measure. However, tests used to assess performance and symptom validity do not directly establish the construct validity of neuropsychological scores. Instead, they indicate when responses (scores) are so improbable that the entire test profile is unlikely to be accurate or interpretable [3]. This distinction is important because the term ‘validity’ may possess different meanings depending on the context. Within the broader psychometric literature, validity is generally conceptualized as the degree to which there is an accumulation of evidence to support interpretation and use of scores [4]. Theoretically, performance and symptom validity may be one source of evidence regarding the interpretation and use of scores in this context; but that is not typically how performance and symptom validity have been used throughout neuropsychology literature. Although the conceptual understanding of validity still requires clarification, there is a clearer understanding of what performance and symptom validity *assessment* entails and how it is defined.

Performance and symptom validity assessment refers to the implementation and interpretation of evidence-based

tools to evaluate the accuracy and interpretability of scores obtained during an evaluation. Performance- and symptom-based scores are assessed separately because they serve distinct purposes. Performance validity tests (PVTs) help determine whether performance-based scores provide a valid estimate of one's actual cognitive abilities [1]. Symptom validity tests (SVTs) help determine whether scores from self-report measures or structured/semi-structured interviews are a valid reflection of one's symptomatic experiences [1].

Defining Invalidity Outcomes and Sources

When multiple PVTs/SVTs are failed, and the resulting scores are thought to reflect improbable patterns of responding (per known base rates), the test profile is deemed invalid and often therefore uninterpretable [5]. “Invalid response set” has been proposed as a more superordinate term to describe a pattern of responding across a test battery indicating that the data are invalid for their intended interpretive purpose [6]. Importantly, these terms represent outcomes, not explanations for why validity issues occur.

Many factors can cause validity issues. When the cause is known, its underlying mechanism should be described (e.g., distraction, low effort). When a factor is empirically associated with invalidity, but its causal role is uncertain (e.g., anxiety, fatigue), it should be described as a possible contributor rather than an established cause. In practice, examiners may have hypotheses about why validity concerns emerge, but insufficient evidence to describe such factors in detail (e.g., distraction, anxiety, etc.). In these cases, examiners may describe the way in which the scores are impacted rather than the suspected causal factors.

As shown in Table 1, these terms are organized by the type of error they introduce. Although terminology varies across contexts, these terms generally refer to distortions between observed scores and the latent constructs those scores are intended to represent [7]. Many theorists have conceptualized these distortions based on their degree of systematicity [7]. Factors that introduce systematic error distort scores in a consistent direction or recognizable pattern, whereas factors that introduce nonsystematic error have more variable and transient effects. This distinction helps clarify how scores are impacted, but it should not be

Table 1 Terminology involving different types of measurement error that compromise performance and symptom validity.

Terminology commonly associated with systematic distortion and measurement error

Response Bias: A systematic tendency to respond inaccurately to clinically meaningful indicators, including cognitive tests, symptom inventories, interviews, or other psychological data. Historically, response bias has referred to responding influenced by factors other than item content [9], but the concept can extend to broader neuropsychological assessment data. It reflects systematic error that may distort estimates of symptoms, abilities, psychological functioning, disability, or capacity [10]. The term is motivationally neutral and does not imply deception.		
Negative Response Biases	Patterns in which test results, symptom reports, or clinical presentation portray the individual as more impaired, symptomatic, or poorly functioning than expected based on broader clinical evidence. Negative response bias refers to the direction of the bias, not its cause or motivation [11, 12]. It should not be used as a synonym for feigning or malingering. The term is sometimes used interchangeably with “negative presentation bias,” although definitions vary. Negative presentation bias may also describe a clinically concerning presentation with evidence less compelling than that required to support malingering [13].	
<i>Intentionality is not required by definition</i>	Overreporting	A form of negative response bias involving excessive endorsement of psychological, cognitive, medical, somatic, or functional problems. It is most appropriate when the person reports more symptoms or difficulties than expected on self-report measures, interviews, or rating scales [12, 14]. The term is descriptive and does not require an inference of intentional distortion. Although sometimes used synonymously with “overstated pathology,” overreporting is more common in the literature [15].
	Symptom Exaggeration	A form of negative response bias in which symptoms that may be genuinely present are described as more frequent, intense, severe, or disabling than broader clinical evidence would support. This term is useful when symptoms are not assumed to be fabricated, but their reported magnitude exceeds reasonable expectation [13].
	Functionality Minimization	A form of negative response bias in which functional abilities, roles, activities, participation, or daily capacities are presented as more limited than expected based on available evidence. This term is most useful when the primary concern is underrepresentation of functional capacity rather than overendorsement of symptoms per se [13].
	Negative Impression Management / Apparent Unfavorable Self-Presentation	A response style in which the individual’s presentation appears to create an impression of greater impairment, distress, or limitation. This may be reflected in test results, self-report, interview behavior, or discrepancies between reported and observed functioning. Negative impression management falls under negative response bias and indicates unfavorable self-presentation, but it does not establish cause or motivation [9, 12, 13]. The phrase “faking bad” should generally be avoided unless intentional distortion is supported.
	Problematic presentation	A broad descriptive term for clinically concerning negative response styles or questionable presentations that do not clearly meet criteria for feigning or malingering. It may be useful when validity concerns are present, but evidence is insufficient to support a more specific conclusion about intentional distortion or external incentive [13, 16].
<i>Intentionality is required by definition</i>	Feigning	Deliberate fabrication or gross exaggeration of symptoms, impairment, or limitations when a specific external motivation has not been established. Feigning is appropriate when intentional symptom distortion is supported, but evidence does not justify concluding that the individual is motivated by identifiable external incentives [12, 15]. It is also commonly used in simulation and experimental malingering paradigms.
	Malingering	Intentional or volitional misrepresentation, fabrication, or gross exaggeration of physical, cognitive, psychological, or functional problems in the context of identifiable external incentives. These may include avoiding work or responsibilities, obtaining compensation, securing accommodations, gaining access to medication, evading prosecution, or reducing legal consequences [12, 13, 17]. Selective concealment or minimization may co-occur with malingering, but should be described separately when it is the primary response pattern.
Positive Response Biases	Patterns in which the individual presents themselves as healthier, less symptomatic, less impaired, or better adjusted than expected based on available evidence. Positive response bias may involve underreporting genuine difficulties, minimizing limitations, or exaggerating positive qualities [9, 12]. The term is motivationally neutral unless additional evidence supports intentional concealment.	
<i>Intentionality is not required by definition</i>	Underreporting / Understatement	A form of positive response bias involving reduced reporting of genuine symptoms, limitations, distress, or impairment. This term is appropriate when the individual appears to report fewer problems than expected, but deliberate concealment has not been established [9, 12].
	Positive impression management	A response style in which the individual’s presentation appears to create an impression of unusually good adjustment, health, emotional stability, or functional capacity. It is used when the response pattern suggests favorable self-presentation, but evidence does not establish intentional concealment [9, 12]. The phrase “faking good” should generally be avoided unless intentional concealment is specifically supported.
<i>Intentionality is required by definition</i>	Dissimulation / dissimulated presentation	Intentional minimization, denial, or concealment of genuine symptoms, problems, or limitations. Dissimulation overlaps with positive response bias, but implies deliberate action and is most appropriate when evidence supports conscious concealment or minimization of difficulties [12, 13].
Neutral Response Biases	Content-independent response tendencies that may distort test, questionnaire, or interview data without consistently shifting presentation in a negative or positive direction. These response styles compromise validity by introducing systematic patterns unrelated to the construct being assessed.	

Table 1 (continued)

Terminology commonly associated with systematic distortion and measurement error

<i>Intentionality is not required by definition</i>	Acquiescence (Yea-Saying)	A content-independent response style involving a tendency to agree with, endorse, or answer “yes,” “true,” or “agree” regardless of item content [9, 18–20]. In symptom validity testing, acquiescence may inflate symptom endorsement when items are phrased symptomatically, or produce inconsistent patterns when logically opposing items are endorsed. It should be understood as a response style, not direct evidence of symptom exaggeration or malingering.
	Extreme Responding	A response style in which individuals disproportionately select endpoint options, such as “strongly agree,” “strongly disagree,” “always,” “never,” or “extremely severe,” regardless of item content [9, 20, 21]. In symptom validity testing, extreme responding may inflate apparent symptom severity, but does not necessarily indicate intentional overreporting.
	Neutral Bias / Moderacy / Mid-point Responding	A content-independent response style involving a tendency to select middle or neutral response options, regardless of whether they accurately reflect the respondent’s experience or belief [9, 20, 22]. In symptom validity testing, moderacy may reduce apparent symptom severity or produce an overly noncommittal profile. It may reflect true ambivalence, low engagement, uncertainty, limited comprehension, or satisficing.
	Fixed Responding / Straight-lining / Long-String Responding	A content-independent response pattern in which the respondent repeatedly selects the same response option across items, regardless of item content. This may reflect carelessness, low engagement, misunderstanding, a response style, or another factor that reduces meaningful engagement with item content [23–25].

Terminology commonly associated with nonsystematic distortion and measurement error

<i>Intentionality is not required by definition</i>	Content Nonresponsivity	Responding to test or questionnaire items without regard to item content. In symptom validity testing, this term is useful when a profile appears invalid because responses do not seem meaningfully connected to the symptom statements presented. This may include random responding, fixed responding, or other careless patterns [24, 26].
	Inconsistent Responding	A pattern in which a respondent provides incompatible answers to items that should be answered similarly, such as semantically similar items, opposite item pairs, or logically related symptoms [24, 25, 27]. It may be detected through paired items, variable response inconsistency indices, synonym or antonym indices, or profile-level inconsistency scales. It may reflect carelessness, confusion, fluctuating attention, reading difficulty, or random responding.
	Random Responding	Arbitrary or haphazard response selection in which answers are not meaningfully related to item content or the respondent’s actual experience [25, 28]. In symptom validity testing, random responding may produce overendorsement or underendorsement depending on item format and chance response patterns. It is a form of response invalidity, but not inherently evidence of malingering.
	Careless Responding / Insufficient Effort Responding in Questionnaire Research	Responses provided with inadequate attention, effort, or motivation to read instructions, interpret item content, and respond accurately [23–25]. This term is often used as an umbrella term for haphazard, inattentive, inconsistent, or random responding in questionnaire and survey research. In symptom validity testing, it may make symptom ratings uninterpretable without implying intentional exaggeration. The phrase “insufficient effort responding” should be distinguished from performance validity terminology because PVT failures should not be reduced to “poor effort” without additional evidence regarding mechanism or cause.
	Inattentive Responding / Participant Inattention	Item responses reflecting insufficient attention to the task, item wording, or response options. Inattention may be intermittent, with careful responses to some items and careless responses to others. It can increase measurement error, reduce statistical power, distort factor structure, and generate spurious findings [29].
	Rapid Responding / Speeding	Completing items or an entire questionnaire so quickly that careful reading and meaningful interpretation are unlikely. Response time is often used as an indirect indicator because some minimum time is required to read, understand, and answer an item [23, 25, 27]. In symptom validity testing, rapid completion may support concerns about careless or inattentive responding, but should usually be interpreted alongside other indicators.
	Satisficing	Providing a response that is merely acceptable or cognitively easy rather than carefully optimized. Satisficing occurs when respondents reduce the cognitive effort required to answer survey questions. In symptom validity testing, it may appear as excessive midpoint responding, acquiescence, straight-lining, or shallow engagement with symptom items [22].

PVT performance validity test, Systematic distortion refers to a consistent or recognizable pattern of measurement error, whereas nonsystematic distortion refers to variable, random distortion. Neither of these terms implies intentionality. Terms such as feigning, malingering, and dissimulation require evidence of intentional distortion. Other terms describe response patterns and should not be interpreted as evidence of intentionality by themselves. Some terms, particularly negative presentation bias, problematic presentation, functionality minimization, and overstated pathology, are emerging or context-specific and should be defined when used rather than assumed to have field-wide consensus

applied rigidly because many factors can cause both forms of error. For this reason, many measurement scholars use broader terms, like “construct-irrelevant variance” or “non-classical error,” to describe error variance from factors unrelated to the test construct [8].

Operationally-defined terms that cause systematic error typically focus on the examinee-specific response biases or distortions [10]. However, systematic error can also arise from other sources, including test-specific content, scoring/administration errors, normative mismatch, sensorimotor interference, or broader contextual influences. For example, limited English proficiency may systematically depress verbally-mediated scores if the tests were designed for English-speaking individuals [30, 31]. Nevertheless, most terminology in Table 1 falls within the purview of negative or positive response biases. These biases are further distinguished based on whether there is evidence to suggest intentionally biased responding, which is often determined by the score distortion severity and/or presence of an incentive. However, it is not always possible to determine if someone’s response bias is deliberate, even if there is a strong degree of systematic error across scores. For example, patients with fixed illness beliefs, poor insight into their symptoms, and maladaptive response styles may produce systematic distortions in symptom ratings but without deliberation [32]. Some research even provides theories rooted in computational neuroscience as to why patients with functional neurological disorder may systematically underperform on testing without deliberation [33].

Nonsystematic sources of error are also important, but there are fewer operationally-defined terms to describe them (Table 1). With regard to performance validity, nonsystematic error may be caused by factors that acutely and inconsistently interfere with attention/engagement, including active psychosis, acute encephalopathy, severe fatigue/pain, misunderstanding task instructions, and medication effects, among others [34–36]. With regard to symptom validity, boredom, hastiness, low reading abilities, severe emotional distress, or severe fatigue may cause nonsystematic error [37, 38]. Most definitions in Table 1 apply to symptom validity issues, and although most of these examples can impact validity, they may not always be significant enough to cause multiple validity test failures.

Existing research on the causes of validity test failures has largely focused on the construct of “effort,” in part because these tests originally emerged from paradigms designed to detect when individuals are attempting to appear impaired and do not fully engage with or exert sufficient effort during testing. Although PVTs/SVTs are often sensitive to effort-related influences – and many factors known to undermine the validity of test scores relate to effort, or more broadly, engagement and conative issues – it is important

to distinguish the intended purpose and target constructs of these tests from the factors that affect them. Said differently, effort may affect the validity of test scores, but validity tests should not be called “effort tests.” Researchers have also moved away from the term “effort testing” because effort varies along a continuum, from strong to very poor [1], whereas PVTs/SVTs detect a more discrete departure from expected performance or symptom reporting [39]. More specifically, PVTs/SVTs detect when scores or response patterns are sufficiently atypical among individuals with genuine neurological impairment to raise concern that the scores may not accurately reflect their abilities or symptoms. So, even though PVTs/SVTs may be affected by effort, clinicians should refer to PVTs/SVTs as indicators of the validity of the scores they are interpreting, rather than as indicators of the amount or quality of effort exerted.

Using Terminology Appropriately

Although terminology related to performance and symptom validity requires further development, many terms are established enough to be used empirically. We encourage scientists and practitioners to use these terms precisely to reduce ambiguity, minimize controversy arising from inconsistent definitions, improve communication across professions involving psychometric testing, and clarify the conditions necessary for valid interpretation of neuropsychological test scores (see Table 2).

Assessing Validity

Validity assessment involves integrating multiple sources of evidence to determine whether neuropsychological scores provide an accurate and interpretable estimate of one’s functioning [5]. PVTs/SVTs are the best tools for this purpose. These psychometric measures elicit noncredible responses, via their appearance and/or detection paradigms, and provide empirical cutoffs that signal when scores are so unusual or extreme that they are ‘improbable’ estimates of one’s actual abilities or experiences [40]. PVTs/SVTs may be freestanding (i.e., ‘standalone’ or ‘dedicated’ measures exclusively developed to assess performance or symptom validity) or embedded (i.e., subscales, indices, or trials contained within clinical tests or questionnaires). Other non-psychometric methods may be used to corroborate PVT/SVT findings, but these methods are supplemental.

Distinguishing Performance Validity and Symptom Validity

Validity of performance- and symptom-based scores should be assessed separately. This distinction is supported by research showing that, among valid-responding examinees,

Table 2 Key problems, solutions, and future directions in validity assessment.

Overarching issue	Problem / gap	Proposed solution
Clarifying What Validity Means		
Conceptual and Terminological Precision	Current terminology often conflates important differences in validity concepts. Terms describing validity outcomes, response patterns, causes, intentionality, and motivation are sometimes used interchangeably.	Use terminology more precisely & describe its association with broader validity theories. Distinguish between the measure, invalidity pattern (e.g., systematic vs. nonsystematic error), and potential cause (e.g., distraction, poor effort, feigning). Clarify how performance and symptom validity relate to broader theories of measurement validity.
Performance versus Symptom Validity	The relationship between performance and symptom validity remains unclear. PVT and SVT findings often diverge, but this may reflect differences in constructs, test characteristics, symptom domains, populations, or referral contexts.	Clarify when and why performance- and symptom-validity findings converge or diverge. Distinguish the validity constructs from the instruments used to measure them, and determine what patterns of convergence/divergence mean for corresponding scores and whether a meaningful shared validity dimension exists.
Improving How Validity Is Assessed and Interpreted		
Test and Cutoff Selection	There is limited guidance regarding which measures and cutoffs to use. Utility of PVTs/SVTs and their cutoffs varies by test characteristics (e.g., face validity, domain specificity, detection paradigm, cognitive or sensorimotor demands), independence from the clinical construct, impairment severity, and population.	Select measures and cutoffs strategically rather than relying on fixed batteries or universal thresholds. Develop methods for identifying combinations of well-validated measures with population-specific cutoffs that maximize incremental value while minimizing unnecessary redundancy.
Administration Strategy	Optimal timing and sequencing remain uncertain. There is little guidance for when to administer tests and how to address validity concerns as they emerge, especially for SVTs.	Develop more continuous and adaptive strategies. Combine free-standing and embedded indicators at critical points during assessments, and use other discrepancy-based methods (e.g., contrast scores) to supplement times PVTs/SVTs are not administered. Develop staged administration strategies in which subsequent PVTs/SVTs are selected according to earlier findings.
Multiple Indicators and Decision Rules	Simple failure-count rules can obscure the complexity of validity findings. Interpretation depends on battery length, indicator quality, failure severity, and statistical dependence.	Utilize multivariate actuarial models as probabilistic indicators to supplement judgement. Develop multivariate rules that consider the number, combination, severity, independence, and quality of passed and failed indicators.
Contextual and Multivariate Base Rates	A PVT/SVT failure does not have the same meaning in every context. Its significance varies by measure, population, referral question, setting, number of indicators administered, and prevalence of invalidity.	Incorporate context-specific and multivariate base rates. Further expand research on base rates and develop practical methods for combining pretest probability, classification accuracy, and multivariate failure patterns when estimating the evidentiary weight of findings
Supplemental Evidence and Integration	PVTs/SVTs provide the strongest validity evidence but do not capture the entire clinical context. Other evidence (e.g., behavior, records, functioning, neurodiagnostic findings) may be informative but vulnerable to bias.	Integrate psychometric findings with carefully evaluated clinical evidence. Determine which supplemental indicators add incremental value and develop more objective methods for quantifying and integrating discrepancies between external data and test scores.
Methodological and Evidentiary Rigor	Extant literature can make classification accuracy appear more certain than it is. Criterion contamination, fallible criterion groups, dependence among indicators, limited generalizability, and forced dichotomization can distort estimates of validity-test performance.	Strengthen validation methods and accommodate uncertainty. Define criterion groups using converging multimethod evidence, evaluate combinations of indicators, account for conditional dependence, validate across populations, and consider graded/indeterminate classifications for conflicting/arbitrary findings.
Expanding What Validity Assessment Can Become		
Information Expansion	Traditional validity assessment methods capture limited data. Validity is sampled only a few times and often reduced to binary interpretations, while potentially informative is discarded.	Digitally capture richer validity-related information. Examine process-based metrics (e.g., response time), biometrics (e.g., eye tracking), ancillary neurodiagnostic data, and ambulatory/ ecological momentary assessment as potential sources of data.
Analytic Innovation	The amount of validity data needed may exceed what clinicians can effectively integrate manually. Traditional approaches may be poorly suited to accommodate multidimensional, nonlinear, and highly interdependent invalidity patterns.	Use advanced analytics to augment clinical interpretation. Evaluate item-response theory/person-fit methods, Bayesian approaches, logistic and other multivariable models, and machine learning to integrate multiple data streams.
Accessibility and Efficiency	Traditional testing limits accessibility, but remote alternatives introduce new sources of error. Device differences, technical disruptions, environmental variability, and reduced examiner control may influence test scores.	Use virtual methods selectively and validate them across conditions. Evaluate measurement properties and cutoffs across devices, settings, administration formats, and varying levels of supervision.
Responsible Implementation and Professional Stewardship	Technology creates new practical, ethical, and interpretive risks. Infrastructure requirements, algorithmic transparency/bias, privacy, test security, inequitable access, and overautomation could undermine validity assessment.	Keep neuropsychologists at the center of technology-enabled validity assessment. Require transparency, ongoing validation, training, privacy/test-security safeguards, and clinical oversight. Technology should automate and augment, not replace, neuropsychological judgment.

PVT performance validity test, *SVT* symptom validity test

self-reported and performance-based scores are minimally correlated [41, 42]. Similar dissociations have been observed among invalid-responding examinees, suggesting that performance and symptom validity assessment serve distinct purposes [3, 43–46].

Although most research suggests that performance and symptom validity tests measure discrete constructs, some studies have found modest associations among certain populations. De Boer et al. [47] identified correlations between these tests when examinees had strong response bias/exaggeration. Boress et al. [48] found that cognitive symptom overreporting and performance invalidity were more correlated in concussion populations, and psychiatric symptom overreporting and performance validity were more correlated in psychiatric populations. Finley et al. [49] found that PVTs and SVTs may capture complementary aspects of validity in dissociative disorder populations. Erdodi [50] and Raasch et al. [51] further suggest that performance and symptom validity associations may depend on the characteristics of the validity tests themselves, including the sensorimotor modality, item content, or paradigm.

Overall, PVTs and SVTs generally assess distinct, though sometimes related, aspects of validity and should not be treated as interchangeable indicators of a single construct. Although performance- and symptom-based scores may share variance in some contexts, evidence for a superordinate validity construct remains limited and may depend on the referral, test characteristics, and specific forms of invalid responding.

Determining the Number of Validity Tests

Validity should be assessed throughout an evaluation using multiple measures [52]. Evaluations with few or no validity measures are generally viewed as lower quality compared to those with multiple validity measures administered [53]. Practice surveys suggest that five PVTs (two freestanding and three embedded) are typical in clinical evaluations, whereas six PVTs (two freestanding and four embedded) are typical in forensic evaluations [53]. Research suggests 5–7 PVTs may provide adequate coverage in most evaluations [5]. Guidance for SVTs is less clear because fewer SVTs exist and many are embedded within lengthy, broadband psychopathology questionnaires [2]. The optimal number of SVTs may depend heavily on the referral, symptom domains assessed, and specific tests, although there is not compelling evidence that domain-specific SVTs are consistently more accurate at detecting invalidity than broader, domain-general overreporting or infrequency indicators [54–57]. Unlike PVTs, in which test scores become progressively and artificially lower as the number of failures increases [58], a similar pattern may not occur as readily

with SVTs unless an examinee is overreporting most or all psychological/behavioral symptoms [2]. Thus, rather than focusing solely on the number of SVTs to administer, it may be more advantageous to begin with a broader sampling of symptom domains (as relevant to the referral), and then select additional SVTs judiciously based on the presenting concerns, psychometric support, and potential incremental value. Domain-matched SVTs may provide clinically relevant information about the content of possible overreporting, but should not be presumed to provide greater classification accuracy than well-validated domain-general approaches [55–57]. To obtain a broad sampling of overreporting, examiners may use comprehensive instruments, such as the Minnesota Multiphasic Personality Inventory-3 [59], which contains five embedded validity scales that assess infrequent reporting of psychiatric, somatic, and cognitive symptoms. Examiners may also consider using SVTs embedded in brief self-report measures, such as those for depression [60, 61], anxiety [61, 62], post-traumatic stress symptoms [63, 64], post-concussion symptoms [65, 66], or pain [67], which allow for an assessment of overreporting across various domains while offsetting the costs and time burden associated with administering broadband personality inventories. However, SVTs embedded in brief self-report measures are often defined by extreme cut-scores that may be conflated with genuinely severe psychopathology. Examiners should therefore substantiate any potential overreporting flagged by these embedded indicators with additional SVTs or with more targeted and validated SVTs. For example, during an attention-deficit/hyperactivity disorder (ADHD) evaluation, an examiner may first use the extreme cutoffs in the Barkley Adult ADHD Rating Scale-IV to identify any overreporting of ADHD-specific symptoms [68], and if those cutoffs are exceeded, they may then use the SVTs in the Conners' Adult Attention-Deficit/Hyperactivity Rating Scale [69] or the Clinical Assessment of Attention Deficit-Adult [70]. Depending on the findings, the examiner may also consider assessing overreporting of other symptoms that commonly co-occur with ADHD [71]. This example is to merely illustrate one way in which examiners can be more selective and strategic in administering SVTs, rather than relying on fixed rules or heuristics regarding how many SVTs to administer.

Selecting Validity Tests

Examiners should select multiple PVTs/SVTs that are complementary rather than redundant [72]. Highly similar tests may resample the same behavior and add minimal incremental value, whereas highly dissimilar tests may not coherently assess the primary validity construct. Several criteria should guide this selection process.

First, each test should have strong empirical support and classification accuracy, including $\geq 90\%$ specificity and $\geq 50\%$ sensitivity [2]. Second, the face validity of each test should be considered. Some examinees underperform on measures that appear to be genuine cognitive tests while avoiding underperformance on measures that appear obviously easy or as PVTs/SVTs [2]. Administering both free-standing and embedded tests can mitigate this issue. Third, examiners should consider symptom content and domain coverage [50]. PVT domains can be classified as memory- or nonmemory-based. SVT domains may be classified as psychiatric, somatic, or cognitive [5], or more specific symptoms within those domains, such as pain [67] or internalizing psychopathology [60, 62, 73, 74]. SVT domain-content can be further specified by diagnostic-specific symptoms, such as ADHD [68, 70, 75], trauma-related disorders [49, 62, 63, 76], or psychosis [77]. Considering symptom content may help ensure that validity assessment adequately samples the symptomatic concerns most relevant to the referral. There is some research to suggest that examinees may ‘pick and choose’ which symptoms to overreport. For example, research shows that adults with invalid presentations who are undergoing ADHD evaluations are more likely to fail ADHD-specific SVTs than the broader overreporting SVTs in the Minnesota Multiphasic Personality Inventory-2-Restructured Form [78]. However, available evidence does not consistently demonstrate that domain-specific SVTs outperform domain-general infrequency or overreporting approaches [55–57]. For example, meta-analytic findings for the Personality Assessment Inventory indicate that overreporting indicators targeting different symptom content generally show comparable ability to distinguish valid from invalid responding, suggesting substantial overlap in the response tendencies they detect [55, 56]. Thus, domain-specific SVTs may be useful for characterizing the content of potentially invalid symptom reporting, but they should not be assumed to provide greater classification accuracy than broader indicators such as Negative Impression Management or Infrequent Responses. Fourth, the actual sensorimotor and cognitive demands of each test should be considered, as examinees may purposefully or erroneously fail validity tests depending on the basic abilities required to pass them [50]. Fifth, examiners should consider how independent the PVTs/SVTs are from the neuropsychological test constructs being evaluated. Freestanding PVTs are more independent of cognitive ability than embedded PVTs [79]. Freestanding SVTs are also usually more independent of symptom severity than embedded SVTs, though some embedded SVTs may be relatively independent of symptom severity [54]. Sixth, the detection paradigm should be considered. SVT paradigms vary depending on whether they are

content-based or non-content-based. Content-based paradigms may assess negative or positive bias via endorsement of excessive, severe, improbable, inconsistent, overly obvious, or stereotyped symptom patterns [80]. Non-content-based paradigms may assess nonresponding, inconsistent or random responding, fixed responding, acquiescence, negativism, extreme responding, or neutral bias/moderacy [81, 82]. PVT paradigms often depend on skills that are robust to brain injury, such as forced-choice recognition memory following prolonged exposure/encoding [83]. Others may involve developmentally overlearned skills, low-demand attentional consistency, contradictory performances, or utilize extremely low cut-scores.

Choosing Validity Test Cutoffs

Validity test cutoffs may be used if they have been validated in the target population [32]. This is especially important for populations with neurological disease, intellectual disability, severe psychopathology, language differences, or other factors that significantly impact validity testing [35]. For PVTs, severity of cognitive impairment is more relevant than diagnosis alone, as the same cutoff can vary across patients with different neurocognitive diagnoses [84]. Even well-validated freestanding PVTs may require alternate cutoffs in impaired populations [85–87]. This concern is especially relevant for embedded PVTs, which are more commonly affected by genuine impairment [88, 89]. By contrast, much less empirical guidance is available for selecting SVT cutoffs. Giromini and colleagues [90] recently described several methodological recommendations for SVT research that also provide a useful framework for evaluating and selecting evidence-based cutoffs. Chiefly, they recommend prioritizing cut scores supported by criterion-group validation studies in which response-validity classifications are based on multiple converging indicators rather than a single criterion measure. Such criterion groups should be defined using multimethod evidence (e.g., multiple well-validated SVTs, behavioral observations, internal inconsistencies, collateral records, and discrepancies between reported symptoms and documented functioning) to reduce circularity and criterion contamination [91]. When SVTs are used as criterion anchors, preference should be given to measures with robust and replicated empirical support across samples and settings, and their cutoffs should generally demonstrate ≥ 0.90 specificity. Cut-score selection should also be informed by empirical reviews rather than test manuals alone, especially if subsequent research indicates that manual-recommended cutoffs do not achieve adequate specificity [70]. Moreover, because criterion-group and simulation studies have complementary strengths and limitations, optimal SVT cutoffs are best supported by

converging findings across both research designs rather than either approach alone [80].

Deciding Administration Order

Next, practitioners should determine when to administer the tests. This process is often iterative with the selection process, and commonly depends on the referral question, evaluation length, patient presentation, and emerging concerns during testing. Nevertheless, it is generally recommended to administer one PVT in the beginning (to guide subsequent test administration/selection) and then intersperse others throughout the evaluation [92]. There is no specific recommendation for SVTs; however, practitioners may consider the staged approach described above, such as initially administering SVTs embedded within brief self-report measures and then determining when additional SVTs should be administered.

Interpreting Validity Test Failures

When PVTs/SVTs are failed, examiners must determine how likely it is that the failures reflect improbable responding. An actuarial approach is useful at this stage because it can be done quickly in real-time using available scores and helps guide the remainder of testing. However, judgments at this stage are best treated as a starting point rather than a definitive conclusion about the validity of the entire profile [5].

A common approach is to count the number of failures across each indicator. For PVTs, ≥ 2 failures (i.e., any pairwise failure on validity tests) is a common benchmark for invalid performance [5, 93, 94]. If two PVTs each have $\geq 90\%$ specificity and they are statistically independent, failing both of them should occur in $\sim 1\%$ of valid examinees by chance. Generally, a larger number of PVT failures should provide stronger evidence of invalid responding because such failure patterns are increasingly uncommon among valid-performing examinees. Accordingly, requiring more failures before classifying performance as ‘invalid’ generally reduces false-positive rates, even among examinees with significant impairment [84, 94–98]. However, the probability of meeting a multivariate criterion for invalid responding (e.g., ≥ 2 failures) depends on each indicator’s specificity, the number administered, their conditional/statistical dependence, and the rule used to aggregate results [94, 99, 100]. Thus, the 1% estimate for failing two independent PVTs (as noted above) cannot be extended to failure on any two PVTs within a larger battery. Although administering more PVTs creates more opportunities to accumulate failures, it does not necessarily produce a meaningful increase in the observed false-positive rate. For example, Kansner et al. [101] found

that failure on ≥ 2 PVTs remained generally stable when increasing from four PVTs to eight PVTs. Similarly, failure rates in non-malingering samples have generally been lower than estimates based on simple independence assumptions because PVTs commonly share variance [94]. Nevertheless, a fixed two-failure threshold should not be generalized to every combination of PVTs because cumulative specificity also varies according to the indicators selected, their cutoffs, score distributions, and the population being evaluated [97, 100, 102, 103]. Multivariate decision rules should therefore be validated for the approximate number and combination of indicators being interpreted rather than assumed to retain the same specificity across batteries.

Several other caveats apply to the two-failure rule. First, this rule was developed primarily for PVTs rather than SVTs and often assumes ≥ 1 freestanding PVT is failed [104]. Second, it assumes the failed tests are not overly redundant, yet there is no standard for how much redundancy is acceptable in this context [5]. Third, the rule is mostly based on 4–9 PVTs being administered [94, 105–108]. When ≥ 10 embedded PVTs are administered, ≥ 3 failures may be appropriate [102, 103]. Some studies have examined using the proportion rather than the total number of failures, but this approach has not worked consistently well, partly because it may not account for skewed score distributions or the test’s psychometric properties [100]. Finally, when performance falls below chance or well below the cutoff on a forced-choice PVT, one failure may actually provide sufficient evidence of invalidity [109, 110].

For SVTs, failure/elevation count models are less established. Although more SVT elevations are associated with greater likelihood of invalid reporting [111, 112], the dose-response association between SVT elevations and confirmed invalidity varies by population, context, and specific symptoms being reported [40, 78, 113, 114]. Some studies suggest ≥ 2 elevations may indicate symptom invalidity in certain populations, but this threshold may increase false-positive risk in other populations [101, 115]. Using ≥ 3 elevations lowers the false-positive risk but may miss milder or more nuanced invalid responding [111].

Using Base Rates to Contextualize Findings

Importance of Base Rates The interpretive weight of the failures/elevations ultimately depends on the context. Two failures do not render the same meaning in every case. Their significance depends on the base rate of invalidity in the target population/setting, as well as the classification accuracy of the PVT/SVT at the applied cutoff [116]. Base rates help estimate the pretest probability of invalidity, likelihood ratios indicate how much a failure can shift that estimate,

and Bayesian reasoning helps integrate these pieces of information to determine the probability that a failure actually reflects invalid responding [117]. However, this reasoning process becomes increasingly complicated when analyzing multiple, redundant PVTs/SVTs [118]. Although exact posterior probabilities may not be feasible to calculate in every evaluation, base rates should still be considered in some capacity given that the same PVT/SVT score carries different interpretive weight across populations and settings.

Clinical vs Forensic Settings Invalid responding is generally less common in routine clinical evaluations than in forensic-legal evaluations, where external incentives are more common. However, clinical and forensic settings should not be treated as synonymous with the absence or presence of incentives as they may be present in either context. Base rates of performance invalidity among clinical outpatients without known external incentive are generally estimated to be 15%–16% [35, 119], with a pooled PVT failure rate of 10% in studies that excluded patients with potential external incentives [35]. Conversely, performance invalidity rates may exceed 40% in forensic or incentive-laden contexts [120]. Symptom invalidity is less well characterized, although a recent review found SVT elevation rates of 15%–30% across seven studies composed primarily of litigants and Veterans [121].

Referral Question Base rates also vary according to the referral question. For example, in evaluations involving known external incentives, a recent survey found that practitioners estimated probable malingering or symptom exaggeration in 29% of personal-injury cases, 30% of disability cases, and 19% of criminal cases [122]. Thus, the presence of external incentives alone is insufficiently precise for estimating base rates, and broad contextual labels such as “clinical” or “forensic” may obscure meaningful differences across referral types.

Examinee Population Invalidity rates also differ across patient populations. Pooled PVT failure rates are estimated at approximately 33% among individuals with functional seizures, 17% among those with ADHD or mild traumatic brain injury, 13% in multiple sclerosis, 11% in epilepsy, and 6% in Parkinson’s disease [35, 123]. SVT elevation rates likewise vary across populations, with reported rates of 36% across psychiatric samples [124], 22% among Veterans with blast exposure [45], and 13% among cognitively impaired patients with apathy [125].

Type of Validity Measure Base-rate estimates also depend on the measure used and the response domain it assesses. As mentioned, SVTs may assess cognitive, psychiatric,

somatic, or diagnosis-specific complaints, and an elevation on one of these SVTs does not necessarily predict an elevation on another [78, 126–128]. For example, SVT elevation rates in Veteran samples have ranged from 5% to 27% depending on the type of SVT [122, 129]. In another example of individuals seeking psychoeducational evaluations, Harrison et al. [126] reported an average elevation rate of 33% that decreased to 12%–13% when analyses were restricted to certain symptom-specific indicators. PVT failure rates similarly vary according to whether the measure is standalone or embedded, its classification accuracy, and the cutoff applied. For example, Roor et al. [35] found pooled failure rates of 25% on the Word Memory Test, 18% on the Medical Symptom Validity Test or Nonverbal Medical Symptom Validity Test, and 9% on both the Test of Memory Malingering and the hard items of the Victoria Symptom Validity Test. These findings further underscore the need to consider multiple test failures when making determinations about base rates.

Multivariate Rates The number and combination of validity measures administered must also be considered. Even when each measure has a low false-positive rate, administering more measures increases the probability of obtaining ≥ 1 failure. Multivariate base rates estimate the probability of obtaining a particular number of failures across an entire battery and are therefore more informative than the base rate of any single measure. For example, if five independent PVTs each have a 10% false-positive rate, the theoretical probability of failing ≥ 2 by chance is approximately 7%, whereas the probability of failing ≥ 3 is $< 1\%$ [100]. Thus, one isolated failure generally provides weaker evidence of invalidity than multiple converging failures. Larrabee [94] similarly found low false-positive rates for ≥ 2 failures in non-malingering clinical samples. These estimates cannot be applied automatically, however, because validity indicators often share task demands or detection strategies. Failures on two indicators derived from the same test provide less independent evidence than failures across distinct measures. Therefore, correlated indicators should not be treated as fully independent, and their likelihood ratios should not simply be multiplied [94, 118].

Although multivariate base rates are well established for cognitive test batteries, they have been applied less often to symptom rating scales. Shura et al. [130] found that ≥ 4 elevated clinical scales on the Minnesota Multiphasic Personality Inventory-2-Restructured Form occurred in 10.3% of healthy Veterans, 28.0% of Veterans without ADHD, and 21.5% of Veterans with ADHD [131]. More recently, Aita et al. [132] found that ≥ 3 skyline elevations across Personality Assessment Inventory clinical scales occurred in 9.1%

of a genuine military posttraumatic stress disorder sample compared with 58.1% of posttraumatic stress disorder simulators. Multivariate cutoffs yielded sensitivity of 0.65–0.81 and specificity of 0.83–0.98 across comparisons. Although these elevations were on substantive clinical scales rather than traditional validity scales, their rarity may provide supplementary information about potential symptom over-reporting. Similar methods have been developed for other multiscale rating measures, including the Behavior Rating Inventory of Executive Function-Adult Version [133], though this work has primarily established normative multivariate base rates rather than directly detecting symptom invalidity. Accordingly, interpretation should consider the number, independence, severity, and domain of elevations or failures. Whenever available, multivariate base rates should correspond to the number and combination of measures administered in a comparable population.

Using Supplemental Validity Indicators

Although PVTs/SVTs should be prioritized during validity assessment, other sources of evidence can supplement these findings. These methods are not psychometric tests. Rather, they are clinically-driven indications about whether the neuropsychological test scores are coherent, plausible, and consistent with known brain-behavior principles.

One approach is to determine whether the scores “make neuropsychological sense.” This involves assessing discrepancies in scores across similar tests, easier versus harder tests measuring similar constructs, or easier versus harder items within a test [134]. These judgments should be interpreted in light of known multivariate base rates of low scores and intraindividual variability among genuinely impaired populations [135]. The second approach is to evaluate whether scores are consistent with behavior outside the testing environment. For example, extremely low recognition-based memory scores may raise concern for invalidity if the patient provides detailed autobiographical information during the interview, arrives on time, and independently navigates an unfamiliar clinic without difficulty. Examiners may also consider whether the score severity and pattern (i.e., ‘biological severity indexing’) are consistent with the patient’s presenting condition [136]. For example, exceptionally low IQ scores or severely impaired single-word reading would be unusual after a remote concussion/mild traumatic brain injury [131, 137]. Neurodiagnostic findings (e.g., neuroimaging) can substantiate these concerns regarding the discrepancy between observed and expected scores relative to the presenting condition. Several other sources of clinical information may also reveal invalid responding when such information is highly discrepant from the patient’s neuropsychological test scores.

Such information may include symptoms reported by the patient and their informant, medical records, developmental history, occupational and educational status, and physical examination findings. Marked inconsistencies across these sources have long been recognized as relevant to performance and symptom validity assessment [138]. Comparable approaches have also been used in other areas of medicine, such as identifying physical signs that are inconsistent with expected anatomical patterns in patients with low back pain [139]. Such inconsistencies should be interpreted alongside other evidence and do not, by themselves, establish invalid responding.

Integrating Evidence Across Sources

The most defensible validity determinations are based on converging evidence across multiple sources. The number of PVT/SVT failures carries the most weight in this determination, but it should be interpreted alongside the aforementioned supplemental methods. These sources of evidence may then be interpreted within the context of the patient’s diagnostic history, intrapersonal background, and referral context to render a final determination about the validity of the neuropsychological profile.

Limitations to Consider When Assessing Validity

It is important to interpret performance and symptom validity assessment findings in the context of potential methodological limitations in PVT/SVT research. Most concerning, criterion contamination can occur when the validity indicator under study, or a highly overlapping indicator, contributes to assigning examinees to the valid and invalid groups. Because the criterion is itself fallible, this approach can inflate estimates of classification accuracy and demonstrate convergence with the selected criterion measures rather than accuracy relative to an independent validity status [118, 140, 141]. Indeed, classification accuracy and recommended cutoffs vary depending on how single-PVT failures are assigned and which criterion measures are selected [141]. Correlations among indicators also do not, by themselves, mean they are overly redundant or lack incremental validity; the more relevant issue is whether they make the same errors within valid and invalid groups and whether adding an indicator meaningfully improves classification [118]. Moreover, validating each PVT/SVT separately does not necessarily establish the validity of interpreting them together. Combining likelihood ratios from separately validated PVTs/SVTs may overestimate the probability that invalid responding occurred when the measures are not conditionally independent [118]. Some researchers propose that an ‘indeterminate’ classification may be more defensible

than forcing a binary valid-versus-invalid conclusion when findings hover near the threshold between valid and invalid or yield conflicting results [50, 72].

Finally, PVTs/SVTs should not be assumed to function uniformly in populations whose cognitive or linguistic limitations may affect the basic abilities required to pass them. These effects are test-specific rather than an inevitable consequence of diagnosis. For instance, adjusted cut-scores are often needed for examinees with severe amnesia on embedded PVTs derived from memory tests to maintain adequate specificity [142]. In dementia evaluations, most PVTs examined by Martin et al. [143] were affected by cognitive impairment severity, and increasingly conservative cutoffs or failure thresholds were needed to preserve specificity as impairment and battery length increased. Low base rates of invalid responding may further limit the positive predictive value of failures in these settings [144]. Limited English proficiency also appears to affect indicators differentially. Ali et al. [145] found that freestanding PVTs were generally robust in samples with limited English proficiency, whereas several verbally-mediated embedded indicators produced false-positive rates of 20%–50%. Findings in intellectual disability are similarly measure-dependent, with some indicators retaining adequate specificity and others requiring closer consideration of test demands and impairment severity [146, 147]. Even populations who exhibit little to no difficulties on standardized testing may produce false positives depending on characteristics and cutoffs of the indicators [148, 149]. SVTs may be similarly vulnerable when administered to patients with marked apathy, alexithymia, schizophrenia, or extensive trauma history, especially if the SVTs rely on extreme cut-scores [54]. Another notable issue within the extant literature is that symptom-invalidity base rates remain difficult to estimate because most studies use different measures, cutoffs, symptom domains, populations, and criteria for defining invalidity [90]. Some researchers report elevations on a single SVT, whereas others require multiple elevations or a more integrated validity determination [2]. Researchers have also sometimes used PVT failure as part of the criterion for symptom invalidity, despite evidence that performance and symptom validity are mostly dissociable constructs [121]. Together, these limitations further underscore the importance of considering the specific abilities required by each validity indicator, validation in comparable populations, impairment severity, and the base rate of invalid responding according to the specific measure, population, and evaluation context.

Future Directions

Many novel methods for validity assessment have been proposed [3]. Yet, very few have been implemented in practice

settings. One major barrier to implementation may be that these methods are time- and labor-intensive when applied to existing paper-and-pencil formats. Finley [150] argued that digital technologies could help overcome this barrier and subsequently improve validity assessment by (1) generating more informative data, (2) leveraging advanced analytics, (3) increasing accessibility, and (4) improving efficiencies (Table 2).

Generating More Informative Data

As Finley [150] described, a key advantage of digital technologies is their ability to capture a greater range and depth of data generated from existing measures. With more data, it becomes easier to examine the multidimensional nature of validity, allowing examiners to discern how validity fluctuates across and within measures. With current methods, capturing large volumes of data is challenging because neuropsychological evaluations are typically limited to only a few PVTs/SVTs that render one dichotomized score [2]. Digitally recording the testing process would allow multiple process-based metrics to be captured across these tests without administering more PVTs/SVTs. Future researchers may wish to determine whether automatically derived metrics (e.g., response-time variability or item-level response patterns) improve classification beyond established PVTs/SVTs and are reliable across devices and testing environments [151, 152].

These metrics could include indices involving response inconsistency or exaggeration magnitude across items of similar difficulty [153]. Another potential process-based metric is response time. Research shows that response latency is helpful for detecting symptom validity issues, like careless responding [154, 155], as well as performance validity issues given that examinees typically cannot feign consistently slow response latencies across all test items [156–160]. Technology can also unobtrusively capture and quantify data across multiple tests, such as severe score dispersion, which may represent another promising validity indicator [161–164].

In addition to PVT/SVT-specific data, digital technologies can record biometric information, such as oculomotor, cardiovascular, or electrodermal responses. These physiological metrics are often associated with cognitive load/effort, which has, in turn, been linked to deceptive responding [165, 166]. As such, these metrics have the potential to capture a broader spectrum of effort during neuropsychological testing. These digitized assays might help identify not only when patients intentionally or unintentionally fail PVTs/SVTs, but also the extent to which they are putting forth maximal effort on testing; this distinction is important as it shifts the motive of validity testing from identifying

when scores are so low that they are improbable estimates of one's abilities, to identifying data that are plausible and likely estimates of the patient's true maximal cognitive ability, which is ultimately what standardized performance-based tests are designed to measure. This change in focus also aligns with emerging work in 'positive neuropsychology' [167]. The use of biometrics is increasing, especially regarding the use of eye tracking and pupillometry to detect invalidity [168–172]. Other physiological markers have also shown promise. Electroencephalogram recordings, for instance, can signal when epileptic activity interferes with performance validity [173, 174]. Exploratory work has also used neuroimaging to examine whether structural and functional brain abnormalities correlate with invalid responding [175–178]. Although biometric measures may provide objective indicators of effort during testing, it still remains unclear how well they can distinguish invalid responding from noise that is produced by innocuous physiological symptoms. For example, eye-tracking and pupillometry findings can be influenced by pain, psychiatric symptoms, or fatigue, which may appear like validity threats but may actually have no bearing on response validity [168, 169, 171, 177].

Ecological momentary assessment and other ambulatory digital methods may also offer promising, though largely untested, applications for validity assessment. These methods collect information through self-report questionnaires, brief cognitive tasks, and passive sensors of everyday behavior [179–181]. As such, there may be opportunities to examine whether discrepancies between ecological momentary assessment data and in-clinic neuropsychological test data are so unusual and unlikely (relative to expected variability and relevant base rates) that they provide additional evidence of invalid responding. However, an important challenge will be distinguishing implausible discrepancies from legitimate variability associated with factors such as fatigue, context, practice effects, device differences, or inconsistent ecological momentary assessment participation [181, 182]. Future research should therefore establish the number of observations needed to obtain reliable estimates, characterize practice effects, and determine how missing or delayed responses, missing sensor data, and changes in devices or measurement settings affect interpretation [183, 184].

Together, these technologies enable the collection of multiple data streams, and even if these data are derived from the same tests, they may not be overly redundant [150]. Regardless, with access to richer data, researchers can evaluate which streams provide unique or complementary information that improves the validity assessment. In a similar vein, these technologies have the potential to help identify the type of error that contributes to the invalid data, and also potentially determine the extent to which a patient

is performing to the best of their ability, rather than simply determining whether performance is so poor that it represents an implausible estimate of their ability.

Leveraging Advanced Analytics

The second advantage of technology is that it facilitates the use of advanced analytic methods, which are needed to make sense of the large data volumes generated via digital testing. First, technologies can provide the software and hardware needed to implement existing multivariable algorithms that have proven effective in validity assessment, including logistic regression, discriminant function analysis, and Bayesian modeling [2, 185–187]. Second, technologies can leverage more novel and promising approaches, including item response theory (IRT) and machine learning [188, 189]. These approaches are well-suited for handling complex data, including large numbers of interrelated variables, nonlinear patterns, and item-level information [190, 191]. This is important for validity assessment, as invalid responding can be highly variable and unpredictable and elicit 'messy' data [103].

These approaches can also improve PVT/SVT development. For example, person-fit statistics, an IRT-based method, may detect more subtle patterns of invalid responding than traditional cut-scores [192, 193]. Rather than signaling invalidity with a cut score, these methods evaluate whether response patterns are consistent with how each test (or multiple tests measuring similar constructs) is intended to function. When the pattern does not 'fit together,' it raises concern that scores may not reflect true ability [194]. Machine learning offers another type of advantage. It can model complex interactions among variables, such as how the number and severity of PVT/SVT failures relate to medical history, litigation status, or other contextual factors, with sophistication that is difficult for clinicians to track manually [189, 195–198]. Although analytic approaches are not technologies in themselves, they become more powerful when paired with digital platforms that aggregate large volumes of complex data. Several other emerging statistical approaches mentioned elsewhere [199] could also be automated via technology.

Future research should determine how to interpret conflicting outputs from multiple statistical validity models and how these models can be integrated into clinical decision-making without relying on automated classifications alone, since they are likely prone to bias [200]. These algorithms may be best used as adjuncts that flag unusual or potentially invalid response patterns for clinical review rather than as standalone determinants of validity [188, 189]. Studies should also examine whether algorithm performance varies across clinical populations and whether demographic,

clinical, or test-specific factors systematically affect classifications [200].

Increasing Accessibility and Efficiency

Digital/web-based testing may also improve access for underserved or geographically restricted communities, though disparities in technology access must be acknowledged [201, 202]. Most SVTs can be administered via a secure link for completion at home, but breaching test security may become a potential issue [203, 204]. Some digitized PVTs are suitable for remote administration (e.g. [204–206]), ; but embedding PVTs within computerized cognitive screeners for face-to-face evaluations is another way to enhance accessibility, as these screeners are commonly used across diverse settings and populations (e.g. [193, 207]), . Embedding PVTs within digital screeners may be especially important when administered in distracting, less-supervised environments (e.g., primary care clinics). Future researchers may wish to examine the conditions under which remote administration meaningfully affects validity-test performance or classification, as virtual assessment can introduce sources of measurement error that are less prominent or more difficult to control during in-person testing. These may include differences in the device used for testing (e.g., computer, tablet, smartphone), technical disruptions (e.g., signal loss, audio or video lag), variability in testing conditions (e.g., background noise, interruptions, or the presence of other individuals), reduced examiner control over the testing environment, and device-related differences in response recording or latency (e.g., slower responses associated with limited familiarity with the device or use of a trackpad rather than a mouse) [208]. Such factors may introduce systematic or nonsystematic error that could affect test scores and, in some cases, increase the likelihood of erroneous validity classifications. Research should therefore determine whether cutoffs established under one administration format remain appropriate across different devices, administration modalities, and testing environments.

Understanding Priorities and Cautions

When technologies digitally record the testing process – or when tools such as eye tracking gather additional data points – several innovations to validity assessment become possible. Examiners can collect a greater range and depth of data, analyze it using more advanced methods, and do so in ways that are more accessible for patients and efficient for clinicians. Although these advances are promising, there remains limited research regarding their efficacy and effectiveness. In addition to resolving the device- and

technology-specific issues described above, several practical, interpretive, ethical, and professional challenges must be addressed before these approaches can be implemented widely [201, 209].

At a practical level, adoption may require substantial technological infrastructure, technical support, clinician training, and financial investment, which may be particularly challenging for smaller or resource-limited practices [210, 211]. Digital measures may also require ongoing psychometric evaluation as hardware, software, and operating systems change, particularly when validity indicators depend on precise behavioral measurements such as response time [152]. Integration with existing testing platforms and electronic health records may create additional technical challenges when digital validity measures rely on data formats, software systems, or computational infrastructure that are not readily compatible with existing clinical systems [210].

Interpretability and clinical oversight represent a separate set of concerns. Proprietary algorithms should be transparent enough for clinicians to understand how validity indicators are derived and to critically evaluate the evidence supporting their interpretation [212]. Likewise, remote, self-administered, or highly automated assessments may reduce opportunities for clinicians to observe behaviors that are important for interpreting apparent invalid performance, including misunderstanding of instructions, distraction, sensorimotor limitations, or technical difficulties [202, 210]. More broadly, automated systems may identify unusual response patterns or generate classifications without accounting for alternative explanations or the clinical context in which those findings occur. Accordingly, clinical interpretation will likely always be necessary when utilizing these technologies.

Methods incorporating biometric, passive-monitoring, or other continuously collected digital data also introduce heightened privacy and data-governance concerns because they may capture sensitive information beyond conventional neuropsychological test scores [180, 213]. Clear consent procedures regarding how these data are collected, analyzed, stored, and used may be especially important in high-stakes evaluations [214]. Questions also remain regarding which forms of digitally derived information (including raw behavioral data, biometric data, algorithmic outputs, and validity classifications) should be incorporated into electronic health records or made available to other referring providers. These considerations intersect with longstanding concerns regarding test security. Remote administration and other digital technologies could increase opportunities for examinees to screen-record, photograph, copy, store, transmit, or otherwise reproduce protected test stimuli and responses beyond what would typically be possible during

an in-person evaluation. Future work should identify methods for limiting unauthorized capture and dissemination of test materials in ways that are consistent with existing professional guidance regarding test security [215].

Equity and generalizability also warrant consideration. Digitally-derived validity indicators may perform differently across individuals with varying levels of technological familiarity, disability, age, language or cultural background, socioeconomic resources, or access to reliable devices and internet connectivity. Similarly, algorithms and biometric indicators developed in narrowly defined samples may not generalize adequately across clinical populations. These concerns reinforce the need for validation studies that evaluate performance across diverse patient groups and testing contexts rather than assuming that findings obtained under one set of technological conditions will generalize broadly [200].

Perhaps most importantly, technology may make validity assessment appear more intuitive, accessible, and convenient than it truly is. Consequently, these methods may be used without appreciation of the fact that validity determinations depend on well-trained clinicians who specialize in understanding brain-behavior relationships. For these reasons, experienced neuropsychologists must remain stewards of validity assessment, ensuring that digital tools are developed, implemented, and interpreted in ways that preserve clinical nuance, empirical rigor, and ethical standards. If used appropriately, digital technologies can complement rather than replace the uniquely human aspects of validity assessment [150].

Author Contributions J.C.F., G.W.K., M.I.L., M.S.P., and J.R.S. all wrote the main manuscript text and reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Human and Animal Rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Competing Interest The authors declare no competing interests.

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